Weekly Macro Outlook
Model-Based Forecasts

US CPI | US PPI | US Retail Sales | UK Labour Market

MIDAS Features, Lagged PCA, Ridge/ElasticNet/XGBoost

Week of May 12–15, 2026

Methodology

Framework

For each target variable, we run a systematic model search: 5 models $\times$ 4 feature sets = 20 combinations, evaluated on a hard train/test split (train: pre-2023, test: 2023–2025). COVID months (2020-02 to 2020-08) excluded from training. Only models passing the overfitting gate (test RMSE $< 1.5\times$ CV RMSE) qualify for inference.

Feature Construction

All features are strictly **lagged** (no contemporaneous information). High-frequency predictors (daily: oil, VIX, yields, FX; weekly: claims) are aggregated to monthly via MIDAS Beta polynomial weights ($\theta_1 = 1, \theta_2 = 5$, K=22 for daily, K=4 for weekly). Same-frequency regressors enter with 1-month lag.

Multicollinearity Treatment

1. **StandardScaler**: all features standardised to zero mean, unit variance
2. **VIF elimination**: iteratively drop highest-VIF feature until all VIF $<$ 10
3. **PCA**: fit on train-only (ex-COVID), extract 3–5 orthogonal factors, **lag by 1 month**

Model Specifications

- **OLS**: unpenalised linear regression (baseline)
- **Ridge**: L2 penalty, λ by LOO-CV — shrinks correlated coefficients uniformly
- **ElasticNet**: L1+L2 mix, automatic feature selection via sparsity
- **XGBoost**: gradient-boosted trees (depth 2–4, L2=2–5, lr=0.03–0.05)

Probabilistic Output

Point forecast: best model refitted on full ex-COVID sample. **Confidence intervals:** Quantile Regression at $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. **Beat/miss:** Ensemble of Ordered Logit (25%) + XGBoost Classifier (50%) + Quantile Regression (25%).

US CPI YoY — Tue May 12 (8:30 ET)

Best Model: ElasticNet + Lagged PCA(5)

$$\widehat{\text{CPI}}_t^{\text{YoY}} = \alpha + \sum_{j=1}^{5} \beta_j \cdot F_{j,t-1} + \gamma \cdot \text{CPI}_{t-1}^{\text{YoY}} + \varepsilon_t$$

Test R^2 = **0.872** (20 models tested, 19/20 passed gate). Classification accuracy: 60%.

Estimated Coefficients (standardised)

Variable	Coef.	Interpretation
CPI_YOY_lag1	$+0.993$	Near unit-root persistence — inflation is highly autocorrelated
PC5_lag1	$+0.151$	Financial conditions factor (VIX/credit-spread loaded)
PC3_lag1	$+0.097$	Energy-price factor (oil-weighted component)
PC2_lag1	-0.100	Yield curve slope — tighter policy dampens future inflation
PC1_lag1	-0.026	Rates level — higher rates reduce demand-pull inflation
PC4_lag1	-0.007	Near-zero: minimal incremental contribution

The dominant predictor is lagged CPI itself ($\beta = 0.993$), reflecting strong inflation persistence from wage-price indexation and administered price contracts. The PCA factors add marginal information: financial stress (PC5) pushes inflation up, while tighter monetary policy (PC2, slope) pushes it down.

Forecast

Previous	Consensus	Point	Median	80% CI	P(miss)	P(beat)
3.3%	3.75%	2.58%	2.58%	[2.19, 2.82]	28.8%	12.2%

Interpretation: The model forecasts 2.58% YoY, well below the 3.75% consensus. The 80% CI [2.19, 2.82] does not contain the consensus. P(inline)=59% dominates. The model relies on lagged CPI (3.3% in March) and sees the energy pass-through as already partially embedded in the base, while the yield curve tightening signal (PC2) dampens forward inflation. *Caveat:* the model uses data through March; it cannot capture intra-April energy price developments that the consensus may incorporate.

US PPI MoM — Wed May 13 (8:30 ET)

Best Model: ElasticNet + VIF-Cleaned Features

Test R^2 = -0.154 (all 20 models passed gate). ElasticNet selected with strong L1 sparsity — only 2 features retained non-zero coefficients:

Variable	Coef.	Interpretation
PPI_MOM_lag1	$+0.433$	Moderate persistence (lower than CPI — PPI mean-reverts faster)
VIX_midas	-0.040	Higher uncertainty → demand destruction → lower producer prices
All others	0.000	Zeroed by L1 penalty (oil, credit spread, EUR/USD not selected)

Prev	Cons	Point	Median	80% CI	P(miss)	P(beat)
+0.5%	+0.5%	+0.96%	+1.56%	[+0.11, +2.64]	10.6%	34.2%

Interpretation: The model forecasts above consensus (0.96% vs 0.5%), driven by the lagged momentum from March's +0.5%. P(beat)=34% outweighs P(miss)=11%. Oil was zeroed by ElasticNet, suggesting that the oil-to-PPI pass-through lag exceeds 1 month in the current regime.

US Retail Sales MoM — Thu May 14 (8:30 ET)

Best Model: Ridge + Lagged PCA(5)

Test R^2 = +0.049. Modest but positive — consumption is driven by a mix of wealth, confidence, and income effects that PCA captures as a composite.

Prev	Cons	Point	Median	80% CI	P(miss)	P(beat)
+1.7%	+0.3%	+0.37%	+0.46%	[−0.67, +1.09]	26.5%	14.3%

Interpretation: The model sees a sharp normalisation from March's +1.7% to +0.37%, broadly in line with consensus. P(inline)=59% dominates. The wide 80% CI reflects the high noise in monthly retail data. The mean-reversion from March's spike is the dominant signal.

UK Labour Market — Wed May 13 (07:00 BST)

Best Model: Ridge + Lagged PCA(5)

Test R^2 = **0.914**. The highest R^2 across all 4 targets. UK unemployment is highly persistent and well-captured by the yield curve regime (PC1/PC2 encode BoE expectations).

Prev	Cons	Point	Median	80% CI	P(miss)	P(beat)
4.9%	4.9%	5.13%	5.09%	[4.87, 5.28]	0.4%	18.3%

Interpretation: The model forecasts 5.13%, above the 4.9% consensus. P(inline)=81% dominates but P(beat)=18% is non-trivial — the model detects upward drift consistent with the rising trend (4.4% → 4.9% over the past year, vacancies at 5-year low). This would signal further labour market deterioration ahead of the June 18 BoE MPC decision.

Cross-Variable Coherence

The four forecasts paint a coherent macro picture:

- **CPI below consensus + PPI above consensus** (→ margin squeeze): CPI at 2.58% vs PPI at 0.96% MoM suggests producer costs are rising faster than consumer prices — a margin squeeze for corporates. This is consistent with the energy cost pass-through still working through the supply chain upstream (PPI) before reaching the consumer (CPI).
- **Retail inline** (→ consumer absorbing shocks): The 0.37% retail forecast is close to consensus, suggesting the consumer is neither collapsing (which would be deflationary) nor overheating (which would add demand-pull pressure). This validates the “soft landing” narrative.
- **UK unemployment drifting higher** (→ BoE dilemma deepens): At 5.13%, UK unemployment is rising while CPI remains elevated (3.3% YoY in March). This is the textbook stagflationary signal: the BoE cannot cut to support employment without risking re-igniting inflation, and cannot hike without deepening the labour market deterioration.
- **Global spillover:** A below-consensus US CPI would be *dovish* for the Fed (supporting rate cuts), which would compress US yields and — via our VAR spillover analysis — reduce pressure on gilt yields (29% of UK Level FEVD from US). This would partially offset the hawkish domestic signal from rising UK unemployment.

Data: FRED (34 series: CPI, PPI, retail sales, UK unemployment, oil, VIX, yields, FX, claims, etc.), NS factors (5 countries, daily). 20 model combinations per target, COVID-excluded training. This document does not constitute investment advice.